

set-01

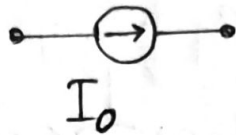
① a

Region

Model

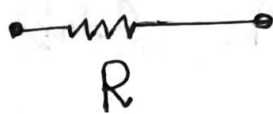
Parameter

AB



$$I_0 = -4 \text{ mA} . (\text{Ans})$$

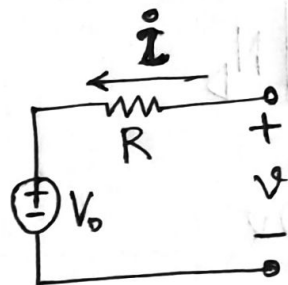
BC



$$R = \frac{1}{m} = \frac{1}{\frac{\Delta y}{\Delta x}} = \frac{1}{\frac{(4 - (-4)) \times 10^{-3}}{2 - (-2)}}$$

$$\hookrightarrow R = 500 \Omega . (\text{Ans})$$

CD



This region can be expressed by, $y = mx + c$; $m = 0.3$

$$\therefore R = \frac{1}{0.3} = 3.33 \Omega$$

Again, this region goes through (2, 4) point.

$$\text{So, } 4 \times 10^{-3} = 0.3 \times 2 + c$$

$$\hookrightarrow c = 0.596 .$$

We know that,

$$\frac{-V_0}{R} = c$$

$$\hookrightarrow V_0 = -R \times c$$

$$\hookrightarrow V_0 = -3.33 \times 0.596$$

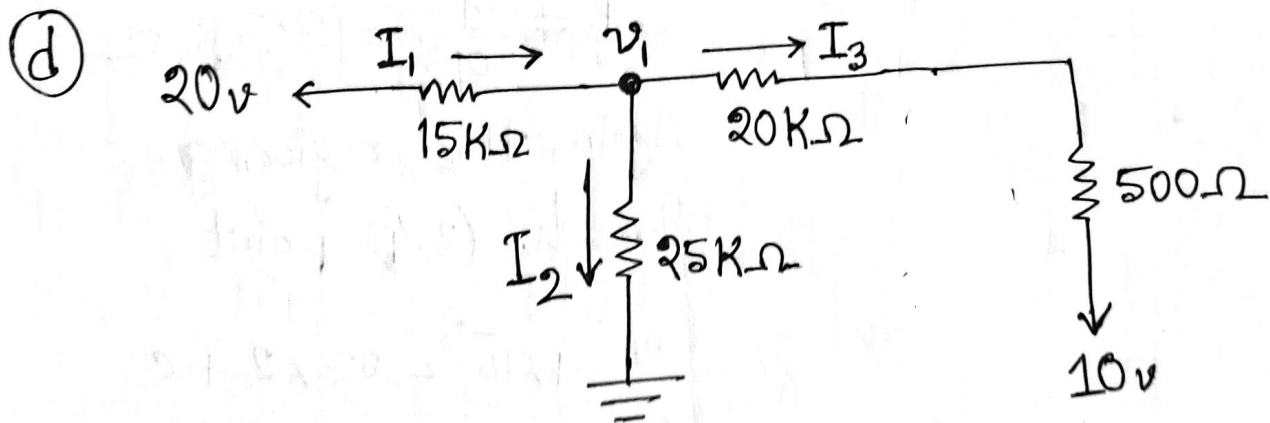
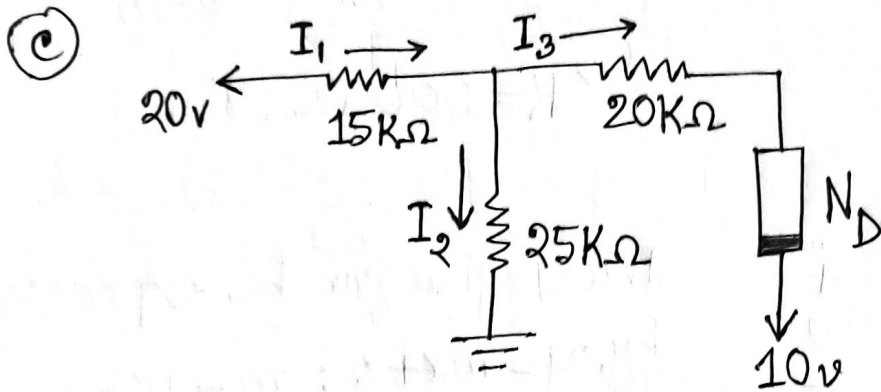
$$\hookrightarrow V_0 = -1.98468 \text{ V} . (\text{Ans})$$

⑥ when, $v_3 = 3v$, the operating region is $\rightarrow CD$
 from part-①, $y = mx + c$

$$\hookrightarrow i_3 = m v_3 + c$$

$$\hookrightarrow i_3 = 0.3 \times 3 + 0.596$$

$$\hookrightarrow i_3 = 1.496 \text{ A. (Ans)}$$



⑨ Applying KCL we get,

$$I_1 = I_2 + I_3$$

$$\hookrightarrow \frac{20 - v_1}{15k\Omega} = \frac{v_1 - 0}{25k\Omega} + \frac{v_1 - 10}{(20 + 0.5)k\Omega} \Rightarrow v_1 = 11.7155v.$$

$$\text{So, } I_1 = \frac{20 - 11.7155}{15k\Omega} = 0.5523 \text{ mA. (Ans)}$$

$$I_2 = \frac{11.7155 - 0}{25 \text{ K}\Omega} = 0.46862 \text{ mA. (Ans)}$$

$$I_3 = \frac{11.7155 - 0}{(20 + 0.5) \text{ K}\Omega} = 0.0837 \text{ mA. (Ans)}$$

② Part-a

① ckt-1: let, the expression be, f_1

$$\text{So, } f_1 = wx. \text{ (Ans)}$$

ckt-2: let, the expression be, f_2

$$\text{So, } f_2 = y + z. \text{ (Ans)}$$

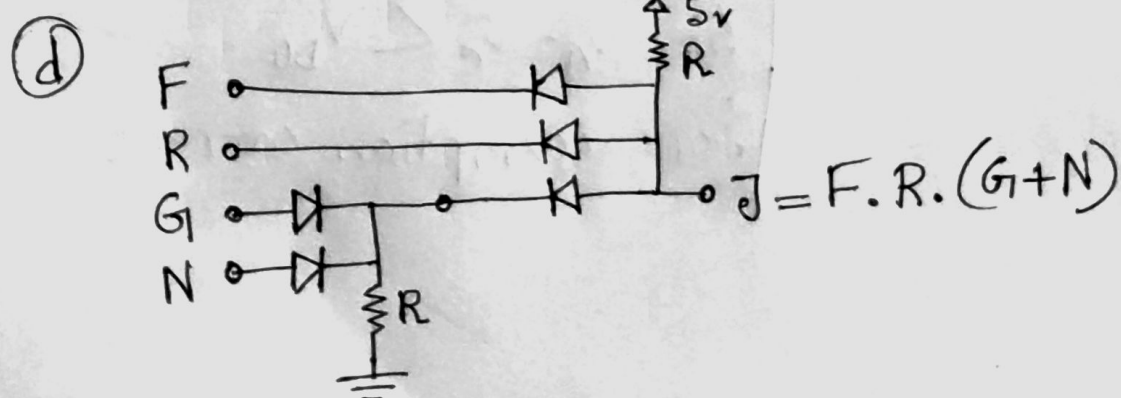
② from the given ckt, we get,

$$f = f_1 + f_2 = wx + y + z. \text{ (Ans)}$$

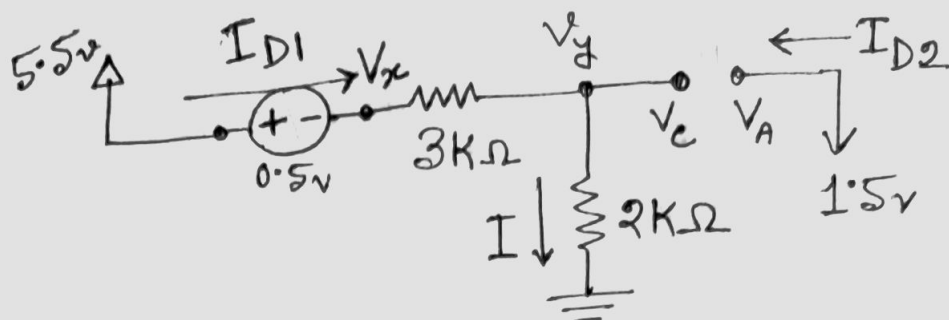
Part-b

① let, the boolean logic function of Juber's algorithm be, J .

$$\therefore J = F \cdot R \cdot (G + N). \text{ (Ans)}$$



③ Let, D_2 off, D_1 ON. So, the circuit will be,



here, $V_x = (5.5 - 0.5)V = 5V$; $I_{D2} = 0$.

$$\therefore I_{D1} = I$$

$$\hookrightarrow \frac{V_x - 0}{(3+2)k\Omega} = I$$

$$\hookrightarrow \frac{5}{5k\Omega} = I$$

$$\hookrightarrow I = 1mA = I_{D1}$$

$$\text{Now, } V_C = V_y = (2k\Omega)(I) = 2k\Omega \times 1mA = 2V$$

for D_1 ,

$$I_{D1} = 1mA > 0$$

$\therefore$ Assumption Correct

Again, for D_2 ,

$$V_A - V_C$$

$$= 1.5 - 2$$

$$= -0.5V$$

$$\text{So, } V_A - V_C < V_{D0}$$

So, assumption correct.

④ a) Purpose: Conversion of AC to DC is done using rectifier.

Operation:

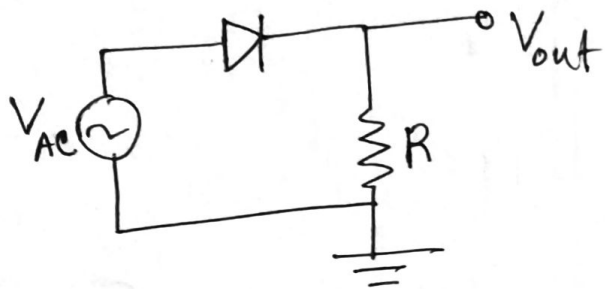


fig: Half-Wave Rectifier

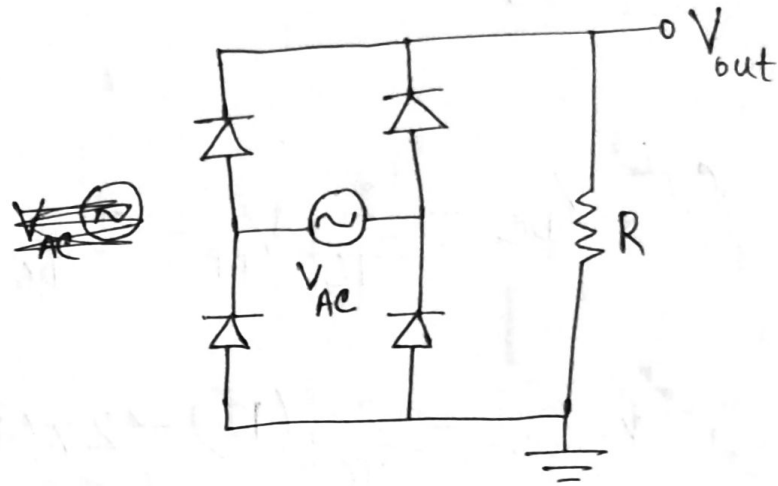


fig: Full-Wave Rectifier

Diode is used to rectify AC. In half-wave rectifier, negative half-cycle of AC voltage gets blocked by the diode. So, we get the positive half-cycle in the output. In full-wave rectifier, both half-cycles can be utilized by connecting the diodes in a smart way which can be seen in the figure above. So we get both half cycles in the output.

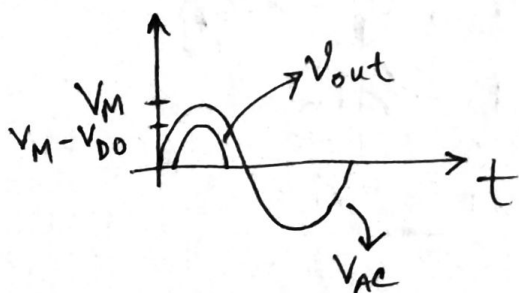


fig: Half-Wave Rectifier

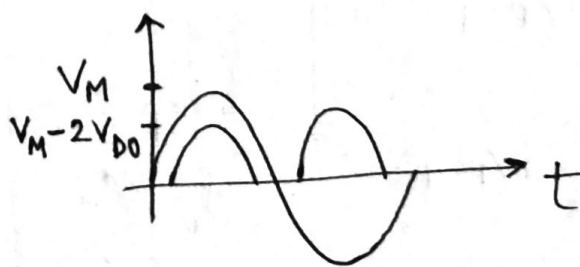
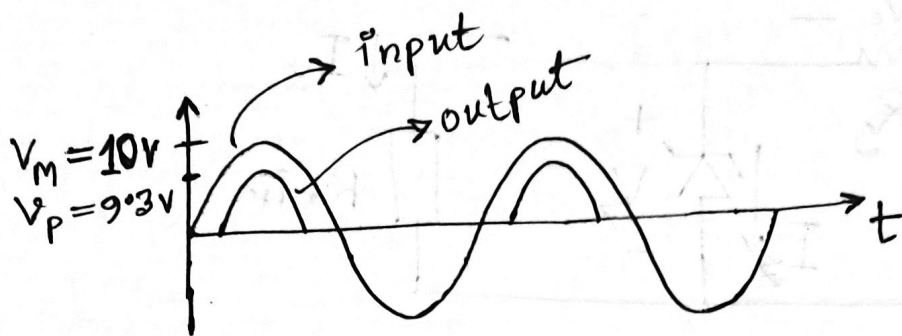


fig: Full-Wave Rectifier

4

6



$$V_P = V_M - V_{D0}$$

$$= (10 - 0.7)V$$

$$= 9.3V$$

c) $V_{DC} = \frac{1}{\pi} V_M - \frac{1}{2} V_{D0}$

$\hookrightarrow V_{DC} = \frac{1}{\pi} \times 10 - \frac{1}{2} \times 0.7$

$\hookrightarrow V_{DC} = 2.833V. (Ans)$

d) $V_{out} = (V_{DC} \pm 0.2)V$

So, $V_{r(p-p)} = V_{out, max} - V_{out, min} = (V_{DC} + 0.2) - (V_{DC} - 0.2)$

$\hookrightarrow V_{r(p-p)} = 0.4V. (Ans)$

Again, $V_{r(p-p)} = \frac{V_P}{f_r RC}$

$\hookrightarrow 0.4 = \frac{9.3}{60 \times 2 \times 10^3 \times C}$

$\hookrightarrow C = 193.75 \mu F. (Ans)$

here,

$V_P = 9.3V$

$f_r = f_i = 60 Hz$

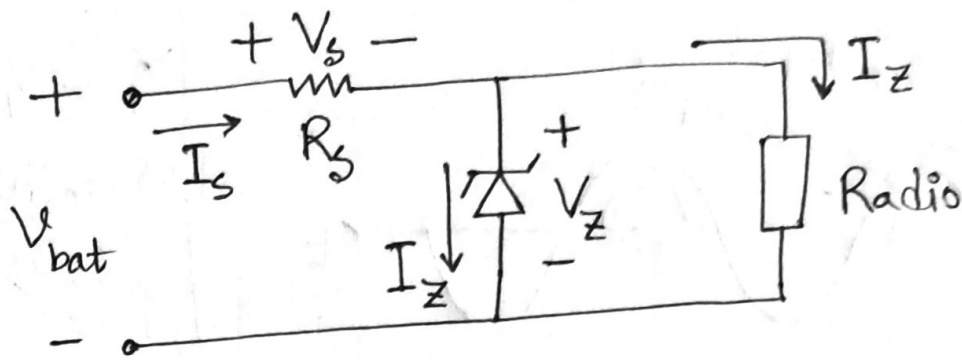
$R = 2K\Omega$

e) $V_{DC} = V_P - \frac{1}{2} V_{r(p-p)}$

$\hookrightarrow V_{DC} = (9.3 - \frac{1}{2} \times 0.4)V$

$\hookrightarrow V_{DC} = 9.1V. (Ans)$

⑤



Given that, $V_{bat} = 11\text{ v} \sim 13.6\text{ v}$

$$I_L = 0 \sim 9\text{ mA}$$

$$V_{z0} = 9\text{ v}, \quad r_z = 0.05\text{ k}\Omega, \quad I_{zk} = 1\text{ mA}$$

(a) Worst case conditions:

$$\textcircled{1} V_{bat(\min)} = 11\text{ v}.$$

$$\textcircled{2} I_z(\min) = I_{zk} = 1\text{ mA}.$$

$$\text{So, } I_z = 1\text{ mA. (Ans)} \quad V_{bat} = 11\text{ v. (Ans)}$$

$$V_z = V_{z0} + I_z r_z$$

$$\hookrightarrow V_z = 9 + 1 \times 10^{-3} \times 0.05 \times 10^3$$

$$\hookrightarrow V_z = 9.05\text{ v. (Ans)}$$

$$I_L = I_{L(\max)} = 9\text{ mA}.$$

⑥ Applying KVL we get, $V_{bat} = V_s + V_z$

$$\hookrightarrow V_s = V_{bat} - V_z$$

$$\hookrightarrow V_s = (11 - 9.05)\text{ v}$$

$$\hookrightarrow V_z = 1.95 \text{ V. (Ans)}$$

Applying KCL we get,

$$I_s = I_z + I_L$$

$$\hookrightarrow I_s = (1+9) \text{ mA}$$

$$\hookrightarrow I_s = 10 \text{ mA. (Ans)}$$

$$\textcircled{c} R_s = \frac{V_z}{I_s} = \frac{1.95}{10 \times 10^{-3}} = 195 \Omega. \text{ (Ans)}$$

Line Regulation,

$$\frac{\Delta V_L}{\Delta V_{bat}} = \frac{r}{r+R}$$

$$= \frac{0.05 \times 10^3}{0.05 \times 10^3 + 195} \text{ V/V}$$

$$= 0.20408 \text{ V/V}$$

$$= 204.08 \text{ mV/V. (Ans)}$$

here,

$$r = r_z = 0.05 \text{ k}\Omega$$

$$R = R_s = 195 \Omega$$

Set-02

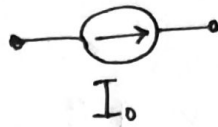
① a

Region

Model

Parameter

AB



$$I_0 = -2 \text{ mA} \text{ (Ans)}$$

BC

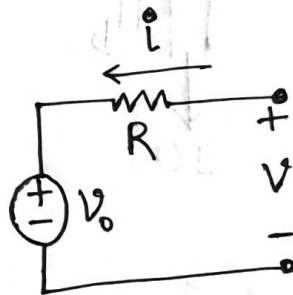


$$R = \frac{1}{m} = \frac{1}{\frac{\Delta y}{\Delta x}}$$

$$\hookrightarrow R = \frac{1}{\frac{(-2-1) \times 10^{-3}}{-4-2}}$$

$$\hookrightarrow R = 1.67 \text{ K}\Omega \text{ (Ans)}$$

CD



This region can be expressed by, $y = mx + c$; $m = 2$

$$\therefore R = \frac{1}{m} = 0.5 \Omega$$

Again, this region goes through (2, 1) point.

$$\text{So, } 1 \times 10^{-3} = 2 \times 1 + c$$

$$\hookrightarrow c = -1.999$$

We know that,

$$-\frac{V_0}{R} = c$$

$$\hookrightarrow V_0 = -R \times c$$

$$\hookrightarrow V_0 = -(0.5) \times (-1.999)$$

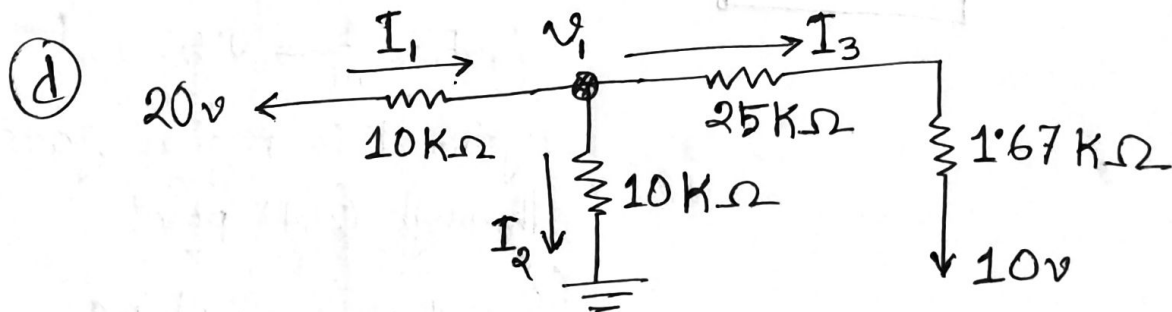
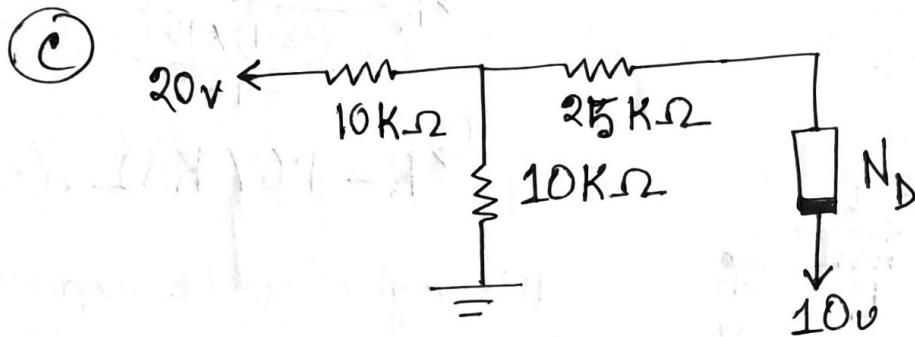
$$\hookrightarrow V_0 = 0.9995 \text{ V} \text{ (Ans)}$$

(b) When, $v_z = 3v$, the operating region is $\rightarrow CD$
 from part-(a), $y = mx + c$

$$\hookrightarrow i_s = m v_z + c$$

$$\hookrightarrow i_s = 2 \times 3 + (-1.999)$$

$$\hookrightarrow i_s = 4.001 \text{ A. (Ans)}$$



(e) Applying KCL we get,

$$I_1 = I_2 + I_3$$

$$\hookrightarrow \frac{20 - v_1}{10K\Omega} = \frac{v_1 - 0}{10K\Omega} + \frac{v_1 - 10}{(25 + 1.67)K\Omega}$$

$$\hookrightarrow v_1 = 10v.$$

$$\text{So, } I_1 = \frac{20 - 10}{10K\Omega} = 1 \text{ mA. } I_2 = \frac{10 - 0}{10K\Omega} = 1 \text{ mA. } I_3 = \frac{10 - 10}{26.67K\Omega} = 0.$$

② Part-a

① ckt-1: let the expression be, f_1 .

$$\text{So, } f_1 = wx. \text{ (Ans)}$$

ckt-2: let the expression be, f_2 .

$$\text{So, } f_2 = y+z. \text{ (Ans)}$$

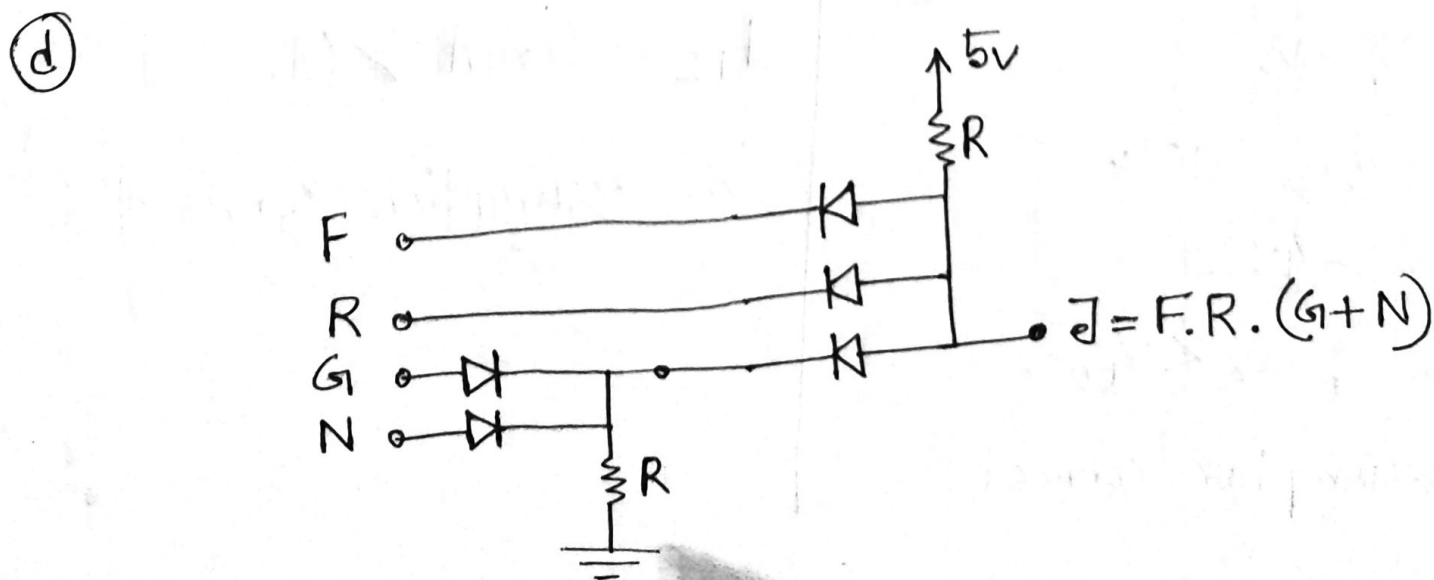
② from the given ckt, we get,

$$f = f_1 f_2 = (wx)(y+z). \text{ (Ans)}$$

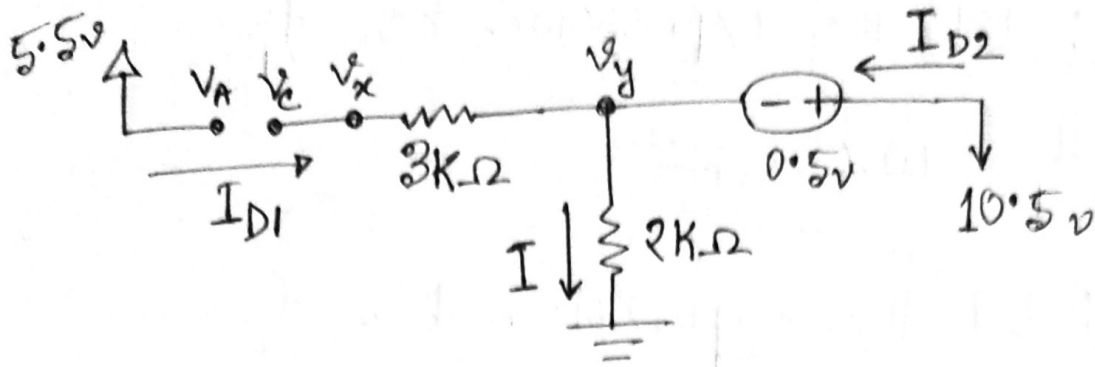
Part-b

① ~~Let~~ let, the boolean logic function of Juber's algorithm be, J .

$$\therefore J = F.R.(G+N). \text{ (Ans)}$$



③ let, D_1 off, D_2 ON. So, the circuit will be,



here, $V_Y = (10.5 - 0.5)V = 10V$; $I_{D1} = 0$.

$$\therefore I = I_{D2}$$

$$\hookrightarrow \frac{V_Y - 0}{2k\Omega} = I_{D2}$$

$$\hookrightarrow I_{D2} = \frac{10 - 0}{2k\Omega}$$

$$\hookrightarrow I_{D2} = 5mA.$$

$$V_C = V_X = V_Y = 10V.$$

Now, for D_1 ,

$$\begin{aligned} & V_A - V_C \\ &= (5.5 - 10)V \\ &= -4.5V. \end{aligned}$$

$$\therefore V_A - V_C < V_{D0},$$

assumption correct

Again, for D_2 ,

$$I_{D2} = 5mA > 0.$$

So, assumption correct.

④ (a) Purpose: Conversion of AC to DC is done using rectifier.

Operation:

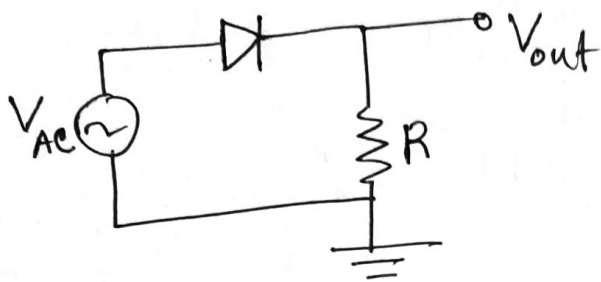


fig: Half-Wave Rectifier

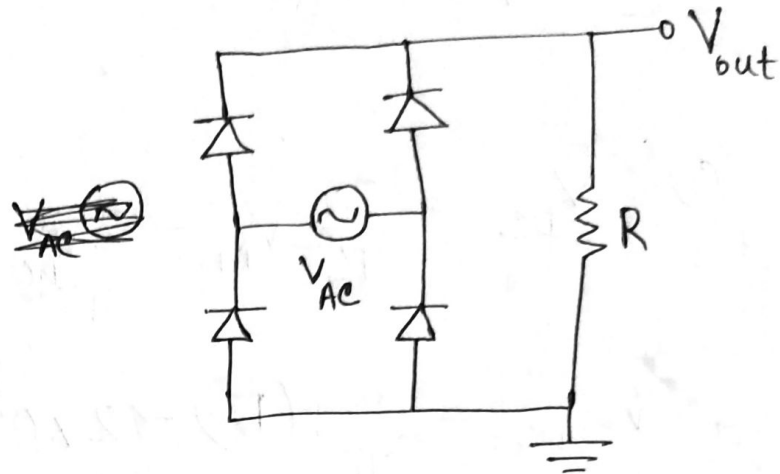


fig: Full-Wave Rectifier

Diode is used to rectify AC. In half-wave rectifier, negative half-cycle of AC voltage gets blocked by the diode. So, we get the positive half-cycle in the output. In full-wave rectifier, both half-cycles can be utilized by connecting the diodes in a smart way which can be seen in the figure above. So we get both half cycles in the output.

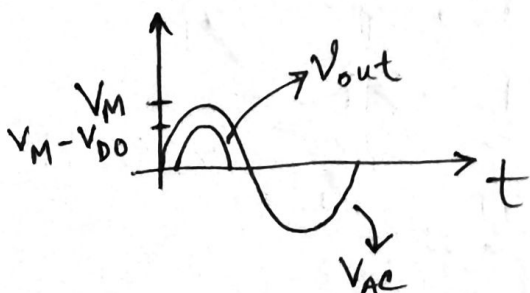


fig: Half-Wave Rectifier

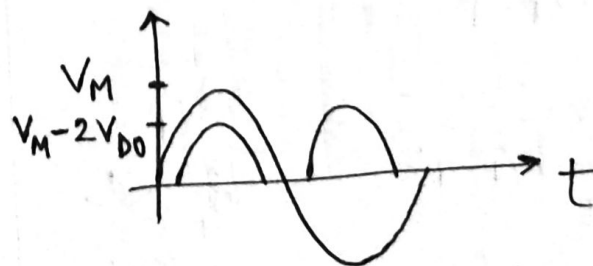
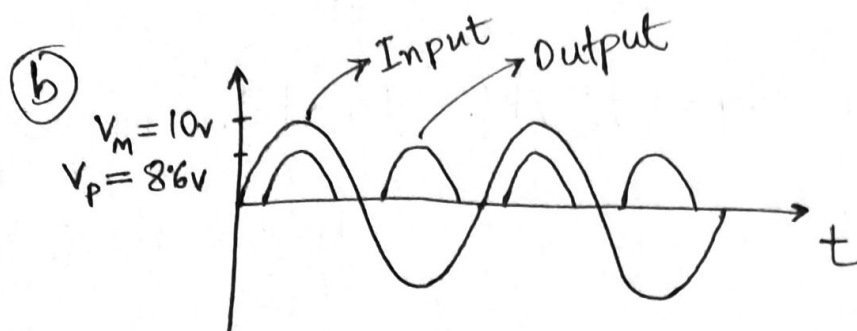


fig: Full-Wave Rectifier



$$V_P = V_M - 2V_{D0}$$

$$= (10 - 2 \times 0.7)\text{V}$$

$$= 8.6\text{V}$$

⑦

$$V_{DC} = \frac{2}{\pi} V_M - 2V_{D0}$$

$$\hookrightarrow V_{DC} = \frac{2}{\pi} (10) - 2 \times 0.7$$

$$\hookrightarrow V_{DC} = 4.9662\text{V. (Ans)}$$

$$(4) (d) V_{out} = (V_{DC} \pm 0.2) v$$

$$\therefore V_{r(p-p)} = V_{out, max} - V_{out, min}$$

$$\hookrightarrow V_{r(p-p)} = (V_{DC} + 0.2) - (V_{DC} - 0.2)$$

$$\hookrightarrow V_{r(p-p)} = 0.4 v. (Ans)$$

$$\text{Again, } V_{r(p-p)} = \frac{V_p}{f_p RC}$$

$$\hookrightarrow 0.4 = \frac{8.6}{2 \times 60 \times 2 \times 10^3 \times C}$$

$$\hookrightarrow C = 89.583 \mu F. (Ans)$$

here,

$$V_p = 8.6 v$$

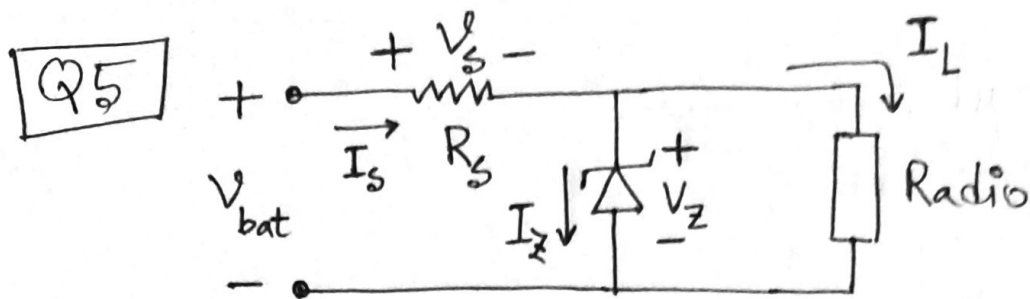
$$f_p = 2f_i = 2 \times 60 \text{ Hz}$$

$$R = 2 \text{ K}\Omega$$

$$(e) V_{DC} = V_p - \frac{1}{2} V_{r(p-p)}$$

$$\hookrightarrow V_{DC} = (8.6 - \frac{1}{2} \times 0.4) v$$

$$\hookrightarrow V_{DC} = 8.4 v. (Ans)$$



Given that, $V_{bat} = 11\text{v} \sim 13.6\text{v}$

$$I_L = 0 \sim 9\text{mA}$$

$$V_{z0} = 9\text{v}, r_z = 0.05\text{K}\Omega, I_{zk} = 1\text{mA}$$

a) Worst case conditions:

$$\textcircled{1} V_{bat(\min)} = 11\text{v}.$$

$$\textcircled{2} I_{z(\min)} = I_{zk} = 1\text{mA}.$$

$$\text{So, } I_z = 1\text{mA} \text{ (Ans)} \quad V_{bat} = 11\text{v} \text{ (Ans)}$$

$$V_z = V_{z0} + I_z r_z$$

$$\hookrightarrow V_z = 9 + 1 \times 10^{-3} \times 0.05 \times 10^3$$

$$\hookrightarrow V_z = 9.05\text{v} \text{ (Ans)}$$

$$I_L = I_{L(\max)} = 9\text{mA} \text{ (Ans)}$$

b) Applying KVL we get, $V_{bat} = V_s + V_z$

$$\hookrightarrow V_s = V_{bat} - V_z$$

$$\hookrightarrow V_s = (11 - 9.05)\text{v}$$

$$\hookrightarrow V_s = 1.95\text{v} \text{ (Ans)}$$

Applying KCL we get,

$$I_s = I_z + I_L$$

$$\hookrightarrow I_s = (1 + 9) \text{ mA}$$

$$\hookrightarrow I_s = 10 \text{ mA. (Ans)}$$

$$\textcircled{c} \quad R_s = \frac{V_s}{I_s} = \frac{1.95}{10 \times 10^{-3}} = 195 \, \Omega. \text{ (Ans)}$$

Load Regulation,

$$\frac{\Delta V_L}{\Delta I_L} = \frac{-rR}{r+R}$$

$$\left| \begin{array}{l} \text{here, } r = r_z = 0.05 \text{ k}\Omega \\ R = R_s = 195 \, \Omega \end{array} \right.$$

$$= \frac{-(0.05 \times 10^3 \times 195)}{(0.05 \times 10^3 + 195)} \text{ mV/mA}$$

$$= -39.796 \text{ mV/mA. (Ans)}$$